# Software Project Management

# What is a software project?

**A temporary endeavour that creates a unique product, service or result**

- **Temporary** — A defined beginning and end
- **Unique** — A specific outcome and context
- **Constrained** — Limited time, people and budget
- **Uncertain** — Requirements and technology may change

# Why software projects struggle

| | |
|---|---|
| **Planning** | Poor estimates, incomplete work definitions, unrealistic deadlines |
| **People** | Unclear roles, weak communication, limited domain knowledge |
| **Change** | Changing requirements, platforms, regulations and priorities |
| **Control** | Weak quality measures, outdated documentation, invisible progress |

# Why software project management matters

- Creates a realistic basis for decisions before development begins
- Coordinates people and technical work across dependencies
- Makes schedule, cost, quality and risk visible
- Supports corrective action when actual progress differs from the plan

# What is Software Project Management (SPM)?

Software Project Management (SPM) is the discipline of planning, organizing, leading, and controlling a software development project so that it is completed on time, within budget, and with the required quality.

It is about managing people, time, cost, scope, and risks involved in building software.

# Key Objectives of Software Project Management

- Deliver software on time, within budget, meeting expectations.
- Coordinate people, processes, and technology.

# Key Activities of SPM

- Project initiation and scope definition
- Planning (tasks, schedule, resources)
- Execution and coordination
- Monitoring and control
- Closure and evaluation

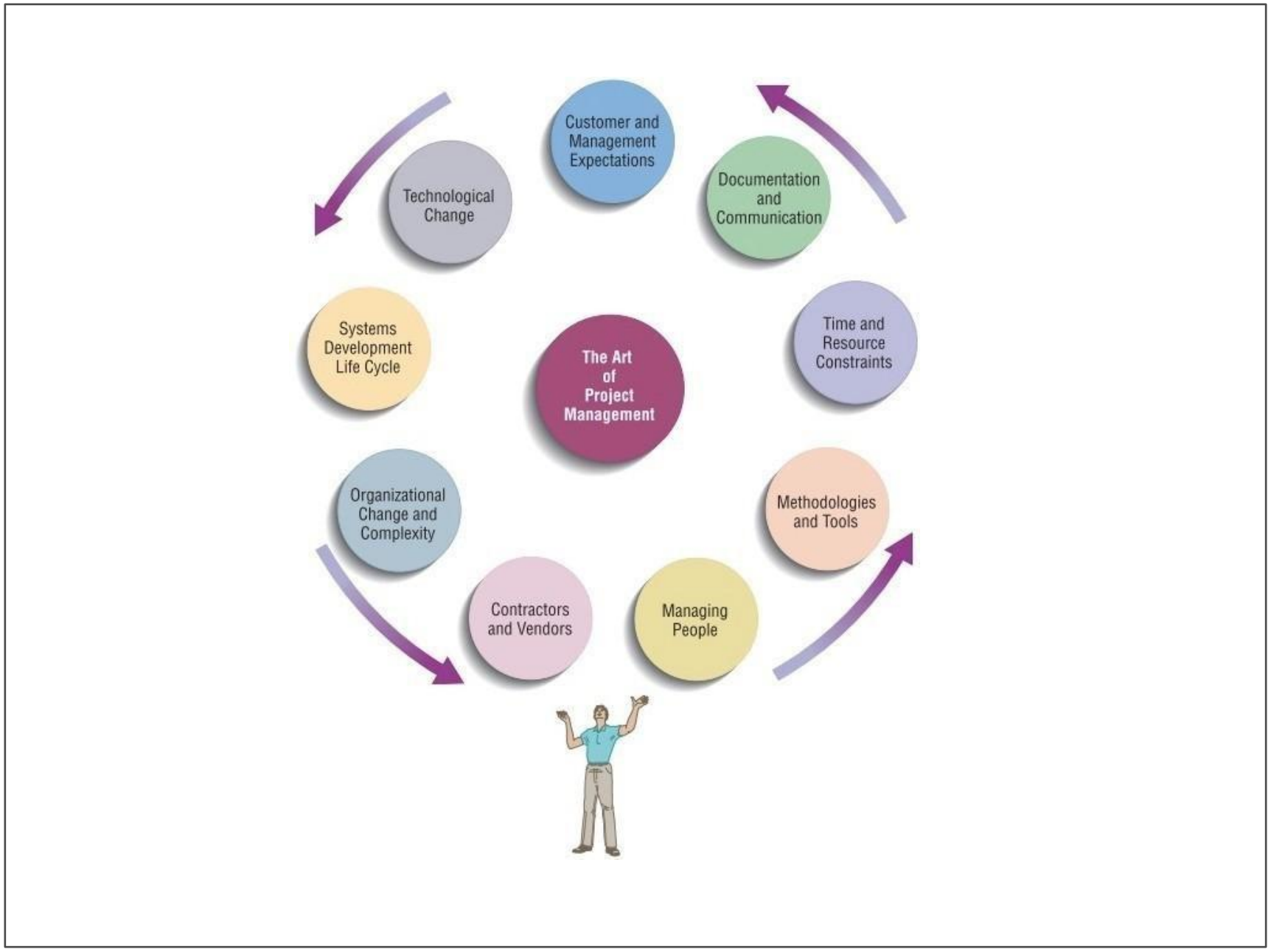
# Project constraints

| | |
|---|---|
| **SCOPE** | What the software must deliver |
| **TIME** | When the work must finish |
| **COST** | People and resources the project can afford |
| **QUALITY** | How well the result must perform |

**A change in one constraint usually changes at least one other constraint**

# Project manager responsibilities

| | |
|---|---|
| **PLAN** | Define scope, work, resources and risks |
| **SCHEDULE** | Sequence tasks and assign dates |
| **MONITOR** | Compare actual progress with the baseline |
| **REPORT** | Communicate status and decisions |


Customer and Management Expectations
Technological Change
Documentation and Communication
Systems Development Life Cycle
The Art of Project Management
Time and Resource Constraints
Organizational Change and Complexity
Methodologies and Tools
Contractors and Vendors
Managing People

# Software project life cycle

1. Initiation
2. **Planning**
3. Execution
4. Monitoring and control
5. Closure

# Project Planning Process

Key components of project planning.

**Scope Management:** Define what's in/out of the project.

- Tools: Work Breakdown Structure (WBS)

**Time Management:** Define tasks, dependencies, and scheduling.

- Tools: Gantt charts, Critical Path Method (CPM)

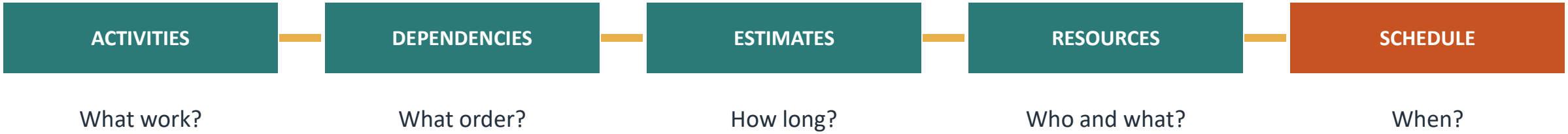
**Cost Estimation:** Resource and budget planning.

- Models: COCOMO

# Project Scheduling

Project Scheduling is the process of listing project activities, determining their sequence, estimating their durations, and assigning start and finish times so that the project can be completed within a specified time frame.

# Project scheduling



# Why?

- Ensure timely completion of the project
- Identify activity dependencies
- Optimize use of time and resources
- Detect potential delays early
- Provide a basis for monitoring and control

# Key Elements

- **Activities** – Tasks required to complete the project
- **Dependencies** – Order in which activities must be performed
- **Duration** – Time required for each activity
- **Start & Finish Times** – Planned schedule for activities
- **Milestones** – Important checkpoints

# Project Activities

Project activities are the basic planning element. Each activity has:

1. A duration in calendar days or months.

2. An effort estimate, which reflects the number of person-days or person-months to complete the work.

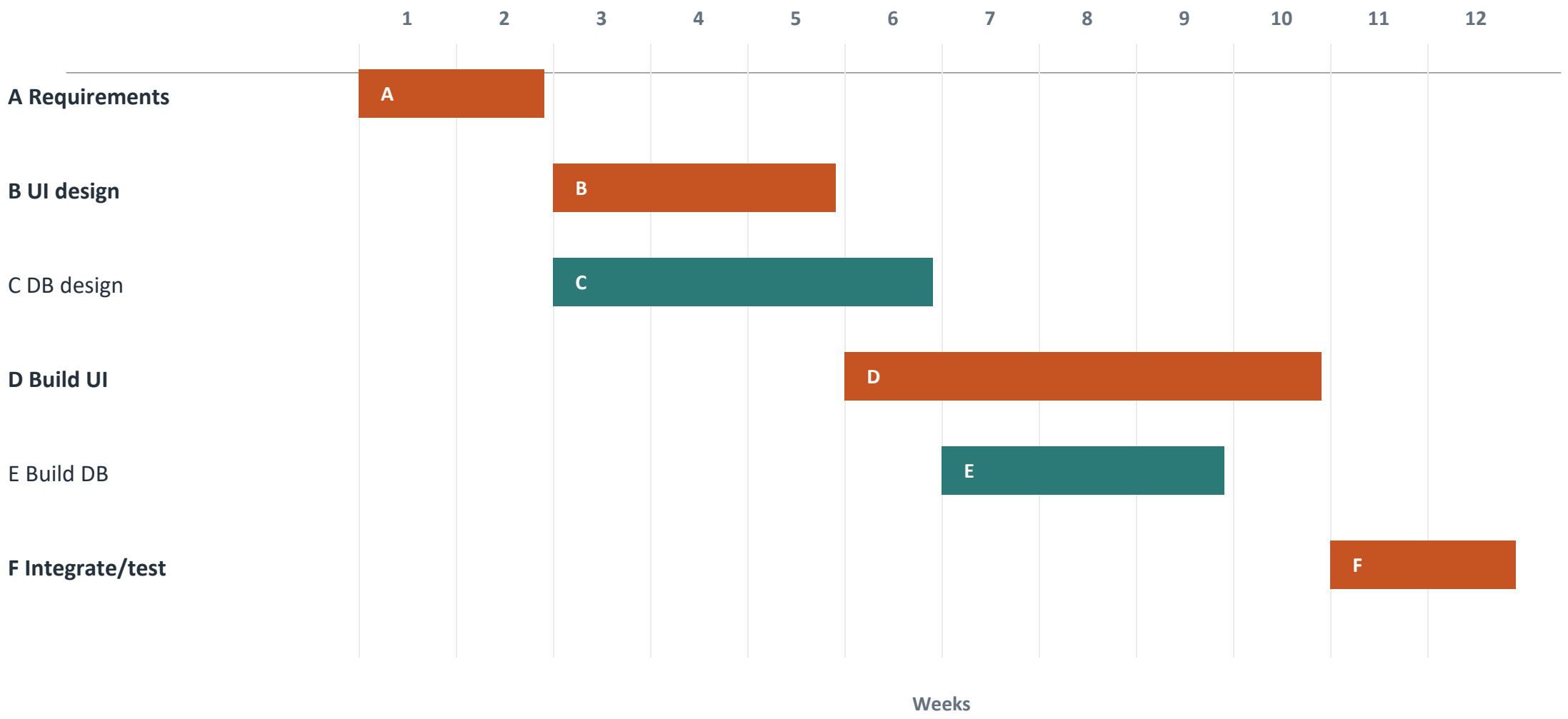
3. A deadline by which the activity should be completed.

4. A defined endpoint. This represents the tangible result of completing the activity. This could be a document, the holding of a review meeting, the successful execution of all tests, etc.

# Techniques

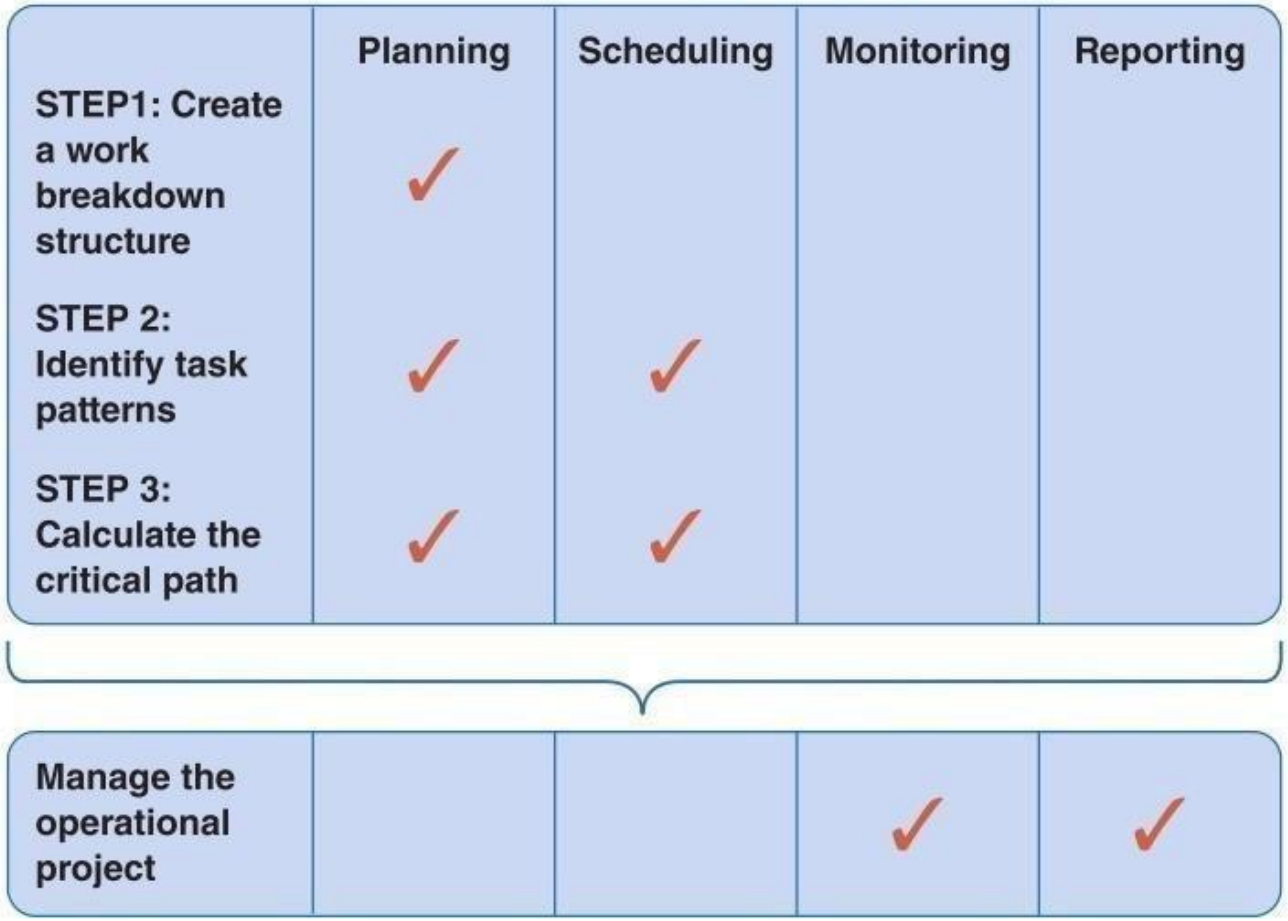
- **Gantt Charts** – Visual timeline of activities
- **CPM (Critical Path Method)** – Identifies longest path and critical activities
- **PERT** – Handles uncertainty in activity durations

# Gantt chart example



# CPM Project Activities

- Create a work breakdown structure
- Identify task patterns
- Calculate the critical path


Planning
Scheduling
Monitoring
Reporting
STEP1: Create a work breakdown structure
STEP 2: Identify task patterns
STEP 3: Calculate the critical path
Manage the operational project

# Work Breakdown Structure (WBS)

A work breakdown structure (WBS) involves breaking a project down into a series of smaller tasks.

# Step1: Create a WBS

**Gantt Chart** - A graph or bar chart with abar for each project activity that shows the passageof time.

**PERT** - Project Evaluation and Review Technique

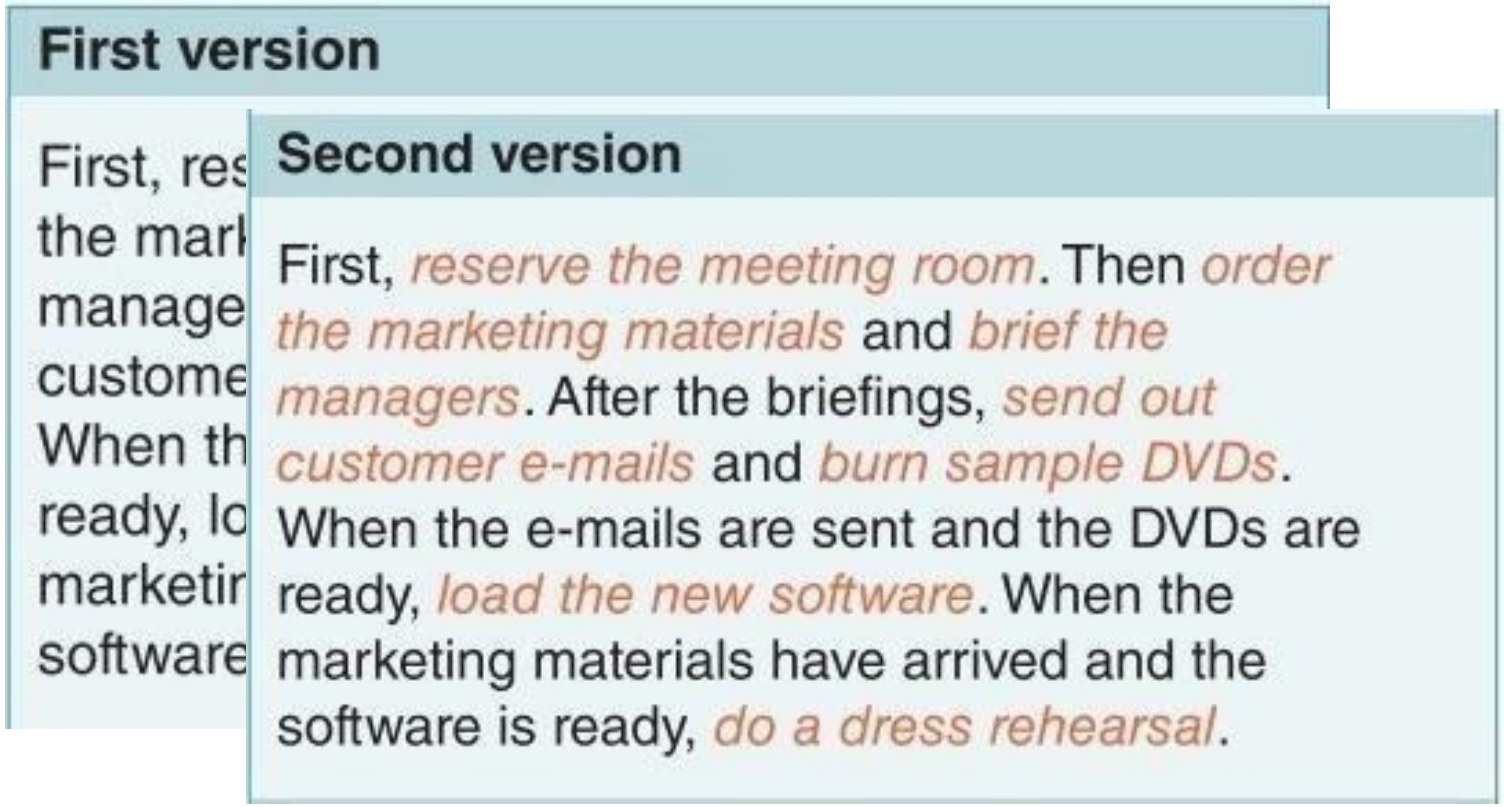
**CPM** - Critical PathMethod

# Identifying Tasks in a Work

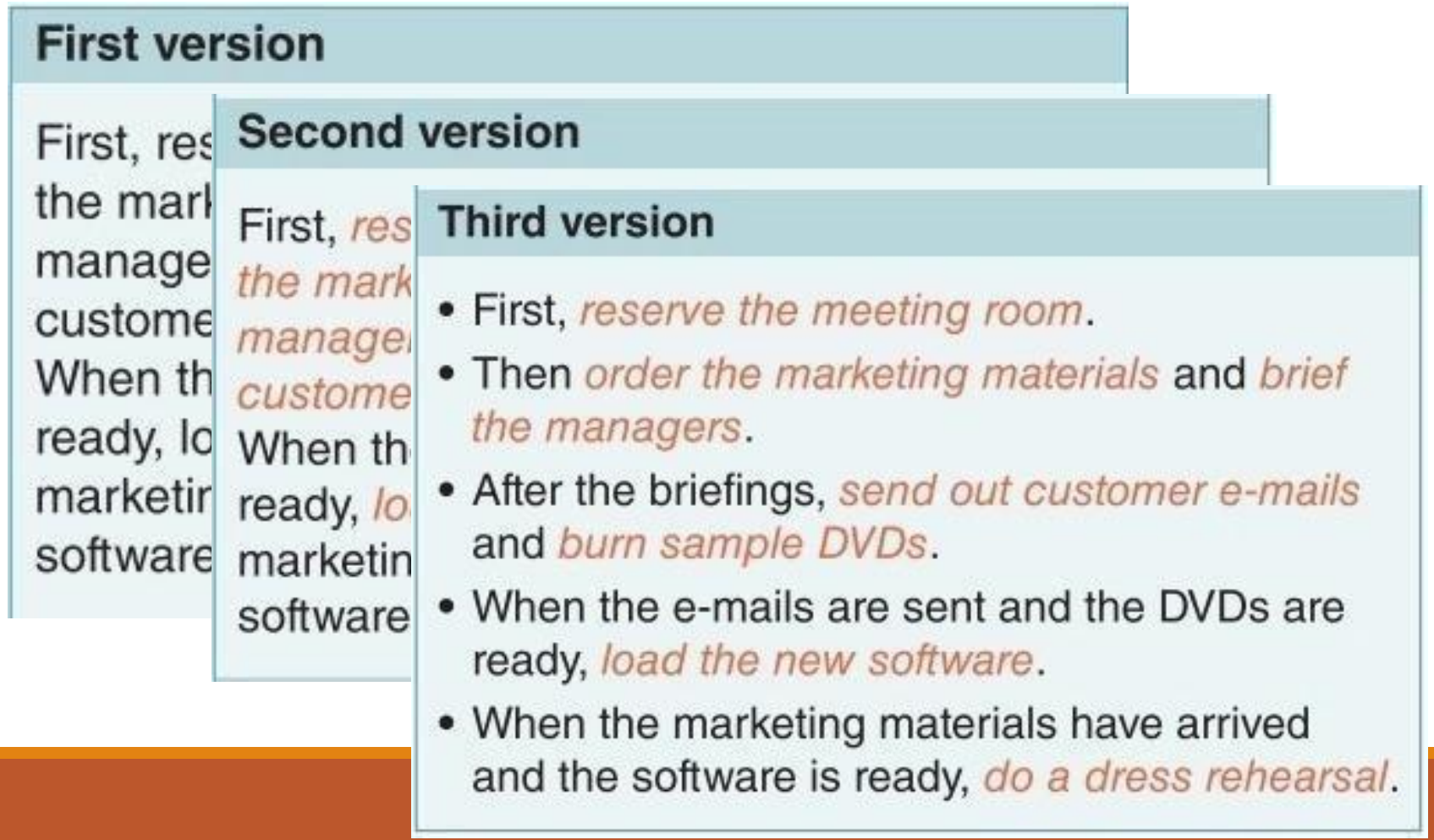
**First version**

First, reserve the meeting room. Then order the marketing materials and brief the managers. After the briefings, send out customer e-mails and burn sample DVDs. When the e-mails are sent and the DVDs are ready, load the new software. When the marketing materials have arrived and the software is ready, do a dress rehearsal.

# Identifying Tasks in a Work



# Identifying Tasks in a Work



# Listing the Tasks

| Task No. | Description | Duration (Days) | Predecessor Tasks |
|---|---|---|---|
| 1 | Reserve the meeting room | | |
| 2 | Order the marketing materials | | |
| 3 | Brief the managers | | |
| 4 | Send out customer e-mails | | |
| 5 | Burn sample DVDs | | |
| 6 | Load the new software | | |
| 7 | Do a dress rehearsal | | |

# Estimating Task Duration

Project managers often use a weighted formula for estimating the duration of each task. The project manager first makes three time estimates for each task: an optimistic, or best-case estimate (B), a probable-case estimate (P), and a pessimistic, or worst-case estimate (W).

The manager then assigns a weight, which is an importance value, to each estimate. The weight can vary, but a common approach is to use a ratio of B = 1, P = 4, and W = 1. The expected task duration is calculated as follows:

$$\frac{(B+4P+W)}{6}$$

# Estimating Task Duration

- A project manager might estimate that a file- conversion task could be completed in as few as 20 days or could take as many as 34 days, but most likely will require 24 days. Using the formula, the expected task duration is 25 days, calculated as follows:

$$\frac{(20+(4*24)+34)}{6} = 25$$

# Estimating Task Duration

| Task | Optimistic Time | Most Likely Time | Pessimistic Time | Expected Time |
|---|---|---|---|---|
| A | 3 | 7 | 11 | |
| B | 5 | 9 | 13 | |
| C | 1 | 2 | 9 | |
| D | 2 | 3 | 16 | |
| E | 2 | 4 | 18 | |
| F | 3 | 4 | 11 | |
| G | 1 | 4 | 7 | |
| H | 3 | 4 | 5 | |
| I | 2 | 4 | 12 | |
| J | 4 | 7 | 9 | |

# Factors Affecting Duration

- Project size
- Human resources
- Experience with similar projects
- Constraints

# Displaying Work Breakdown

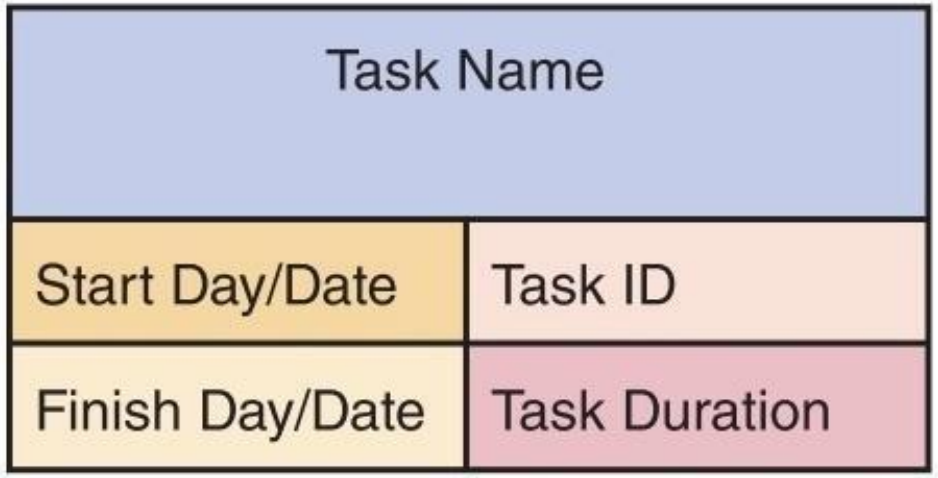
| Task No. | Description | Duration (Days) | Predecessor Tasks |
|---|---|---|---|
| 1 | Reserve the meeting room | 1 | |
| 2 | Order the marketing materials | 9 | |
| 3 | Brief the managers | 2 | |
| 4 | Send out customer e-mails | 3 | |
| 5 | Burn sample DVDs | 3 | |
| 6 | Load the new software | 2 | |
| 7 | Do a dress rehearsal | 1 | |

# Step 2: Identify Task Patterns

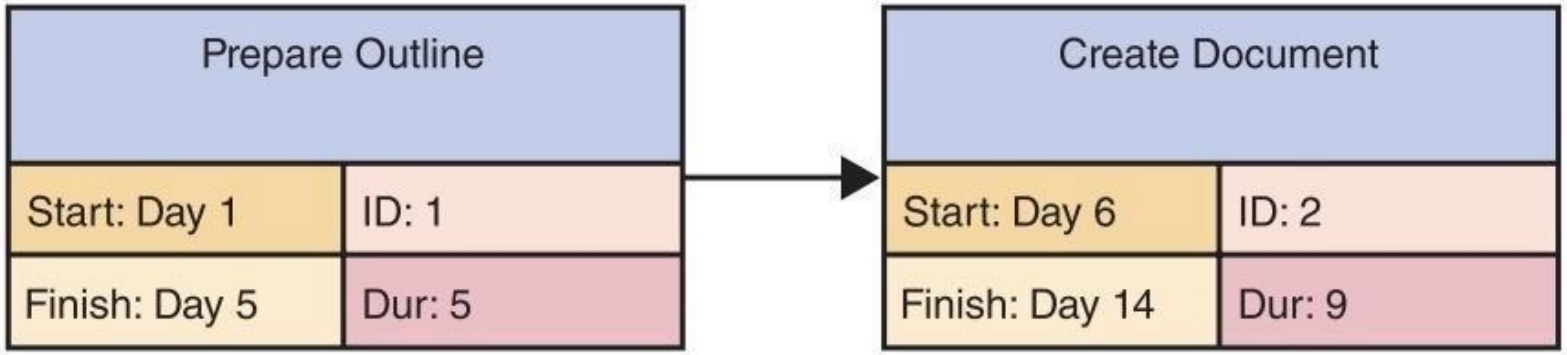
What are Task Patterns?

Tasks in a work breakdown structure must be arranged in a logical sequence called a task pattern.

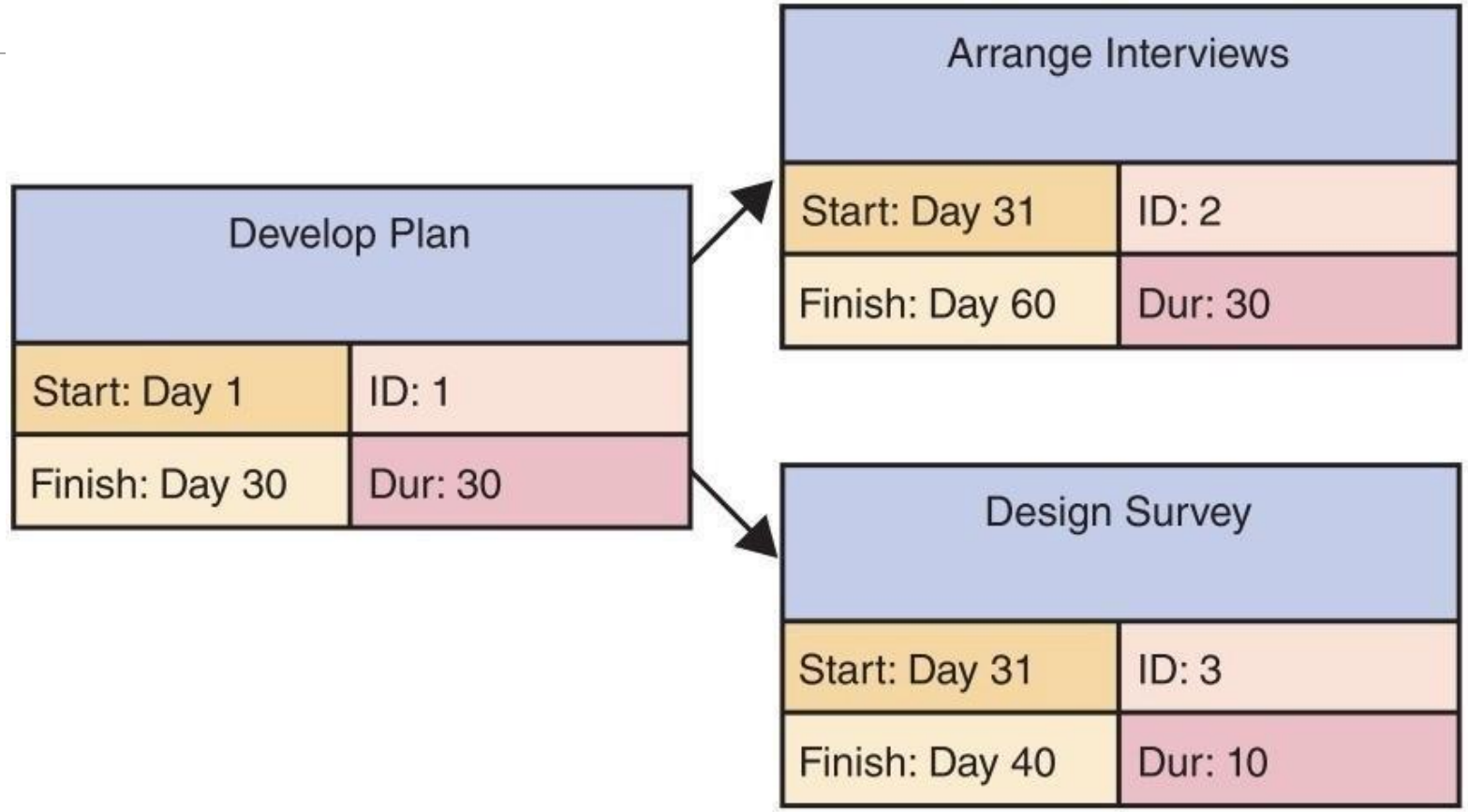
# Task Boxes


Task Name
Start Day/Date
Task ID
Finish Day/Date
Task Duration

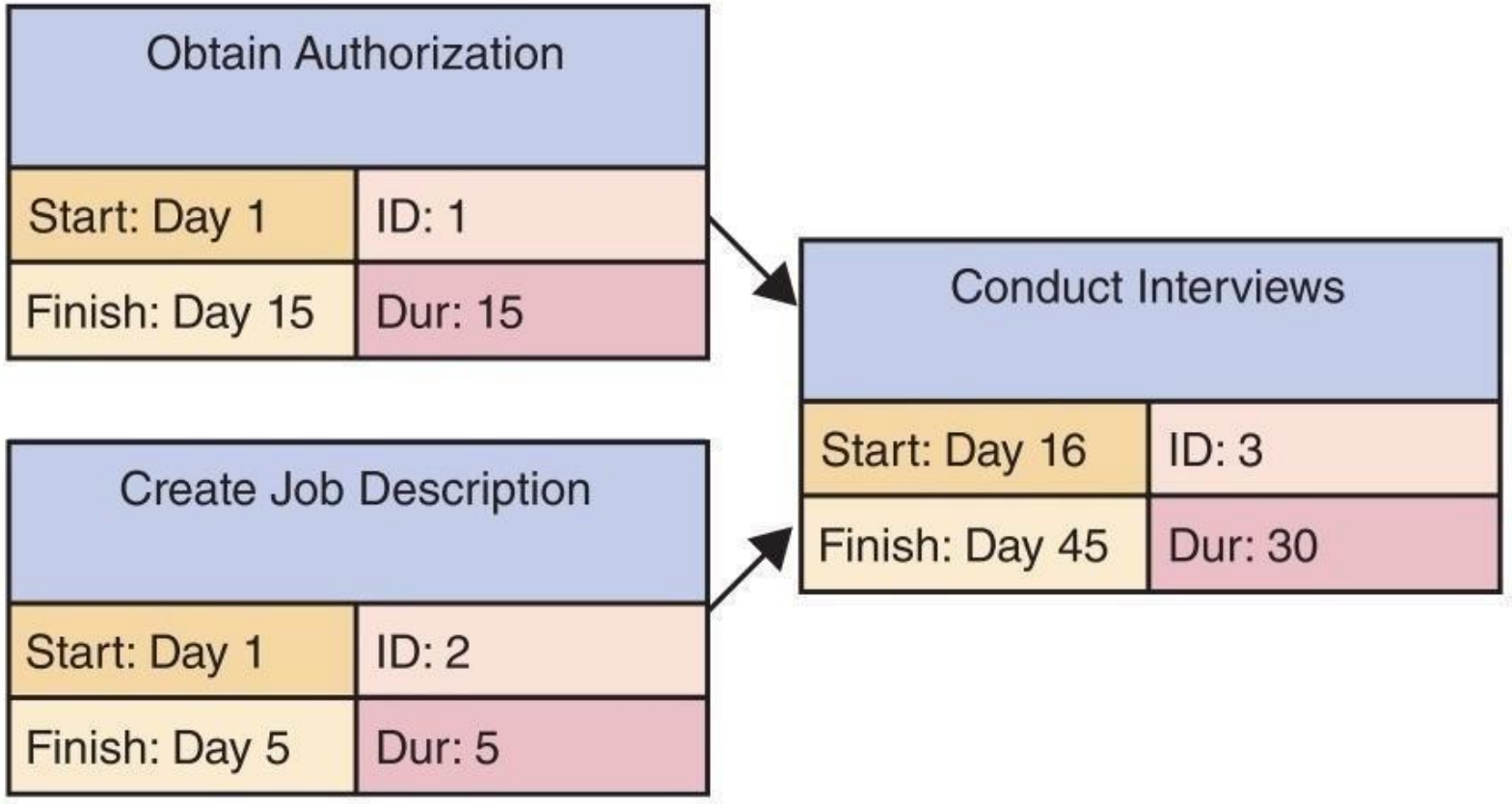
# Dependent Task



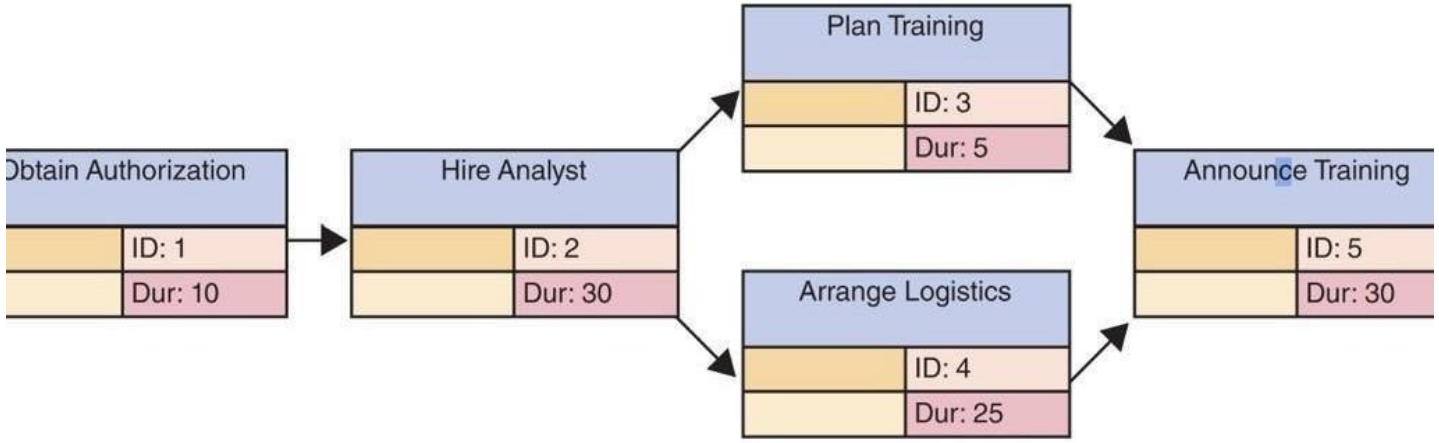
# Multiple Successor Tasks



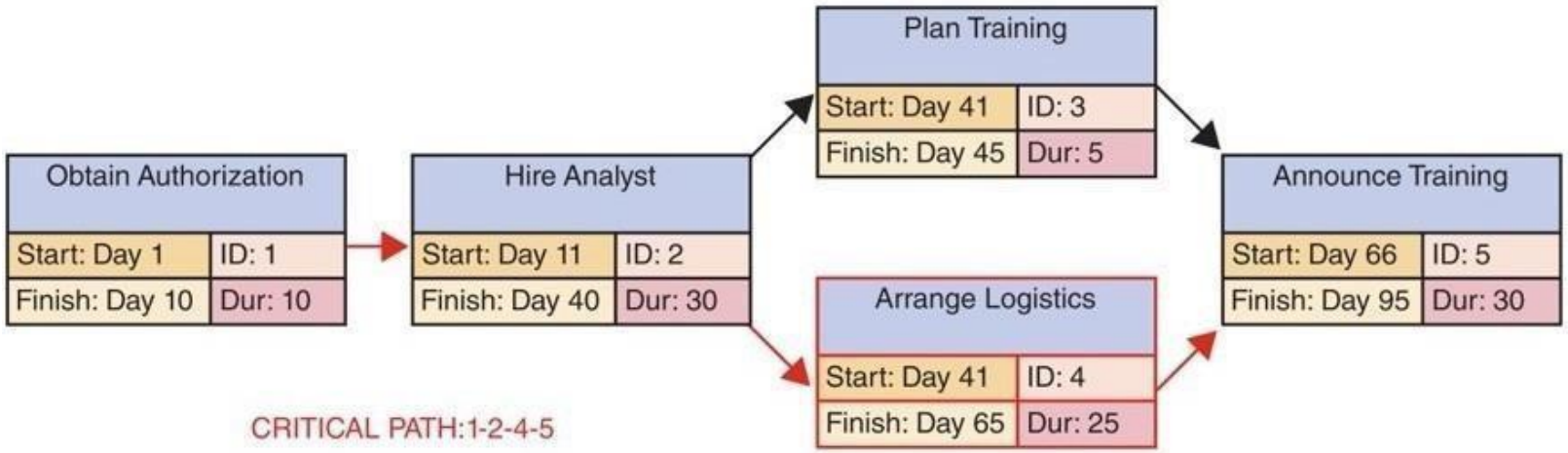
# Multiple Predecessor Tasks



A critical path is a series of tasks which, if delayed, would affect the completion date of the overall project.



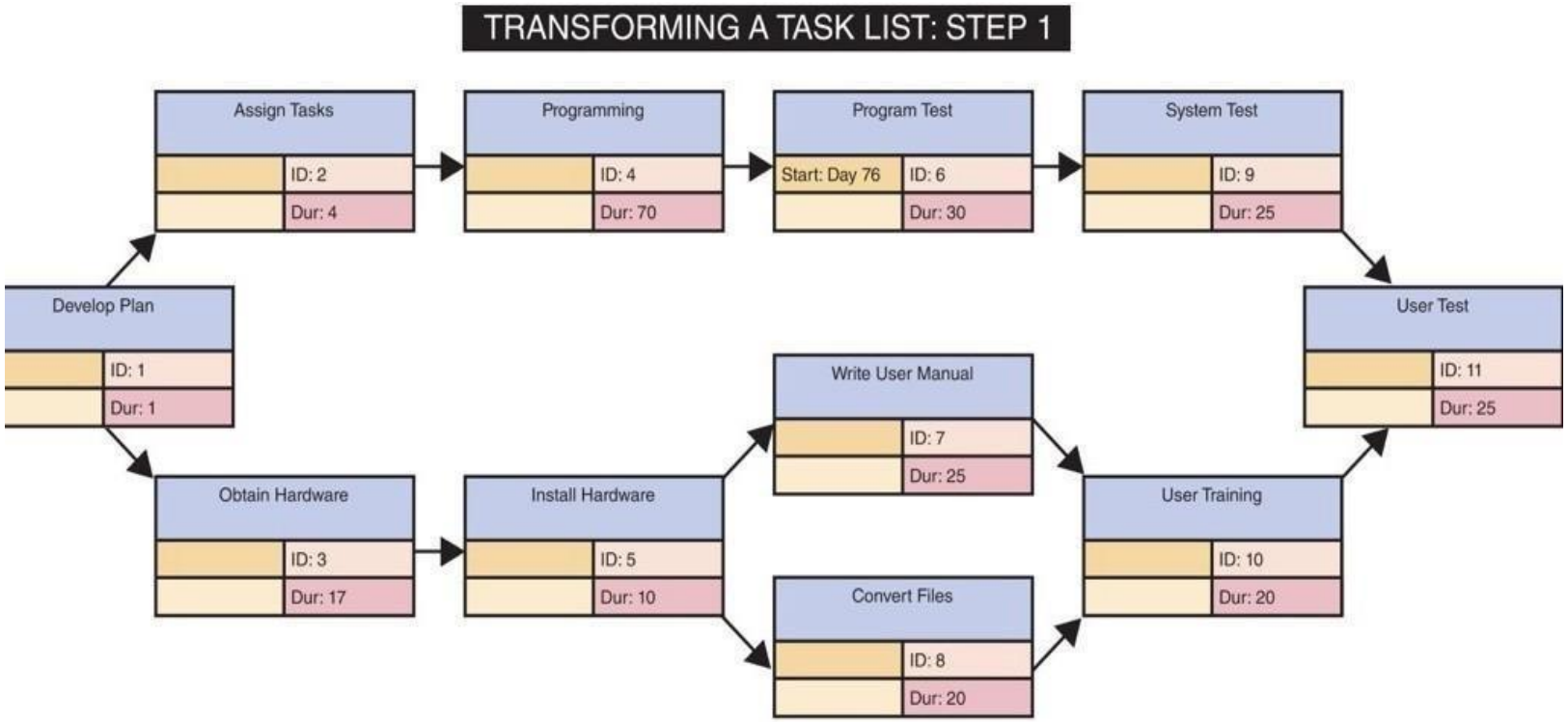
# Step 3: Calculate the Critical Path



# Example

| Task No. | Description | Duration (Days) | Predecessor Tasks |
|---|---|---|---|
| 1 | Develop Plan | 1 | - |
| 2 | Assign Tasks | 4 | 1 |
| 3 | Obtain Hardware | 17 | 1 |
| 4 | Programming | 70 | 2 |
| 5 | Install Hardware | 10 | 3 |
| 6 | Program Test | 30 | 4 |
| 7 | Write User Manual | 25 | 5 |
| 8 | Convert Files | 20 | 5 |
| 9 | System Test | 25 | 6 |
| 10 | User Training | 20 | 7, 8 |
| 11 | User Test | 25 | 9,10 |

# Example


TRANSFORMING A TASK LIST: STEP 1
Assign Tasks
ID: 2
Dur: 4
Programming
ID: 4
Dur: 70
Program Test
Start: Day 76
ID: 6
Dur: 30
System Test
ID: 9
Dur: 25
Develop Plan
ID: 1
Dur: 1
User Test
ID: 11
Dur: 25
Write User Manual
ID: 7
Dur: 25
Obtain Hardware
ID: 3
Dur: 17
Install Hardware
ID: 5
Dur: 10
User Training
ID: 10
Dur: 20
Convert Files
ID: 8
Dur: 20

# Example


TRANSFORMING A TASK LIST: STEP 2
Develop Plan
Start: Day 1
ID: 1
Finish: Day 1
Dur: 1
Assign Tasks
Start: Day 2
ID: 2
Finish: Day 5
Dur: 4
Programming
Start: Day 6
ID: 4
Finish: Day 75
Dur: 70
Program Test
Start: Day 76
ID: 6
Finish: Day 105
Dur: 30
System Test
Start: Day 106
ID: 9
Finish: Day 130
Dur: 25
User Test
Start: Day 131
ID: 11
Finish: Day 155
Dur: 25
Obtain Hardware
Start: Day 2
ID: 3
Finish: Day 18
Dur: 17
Install Hardware
Start: Day 19
ID: 5
Finish: Day 28
Dur: 10
Write User Manual
Start: Day 29
ID: 7
Finish: Day 53
Dur: 25
Convert Files
Start: Day 29
ID: 8
Finish: Day 48
Dur: 20
User Training
Start: Day 54
ID: 10
Finish: Day 73
Dur: 20
CRITICAL PATH:1-2-4-6-9-11

# Tool Support











| Activity No. | Activity | Time (weeks) | Immediate Predecessors |
|---|---|---|---|
| 1 | Collect requirements | 3 | |
| 2 | Analyze processes | 2 | 1 |
| 3 | Analyze data | 2 | 2 |
| 4 | Design processes | 6 | 2 |
| 5 | Design data | 3 | 3 |
| 6 | Design screens | 2 | 3,4 |
| 7 | Design reports | 4 | 4,5 |
| 8 | Program | 5 | 6,7 |
| 9 | Test and Document | 7 | 7 |
| 10 | Install | 2 | 8,9 |

a. Draw a Network diagram for the activities.
b. Calculate the earliest expected completion time.
c. Show the critical path.
d. What would happen if activity 6 were revised to take 6 weeks instead of 2 weeks?